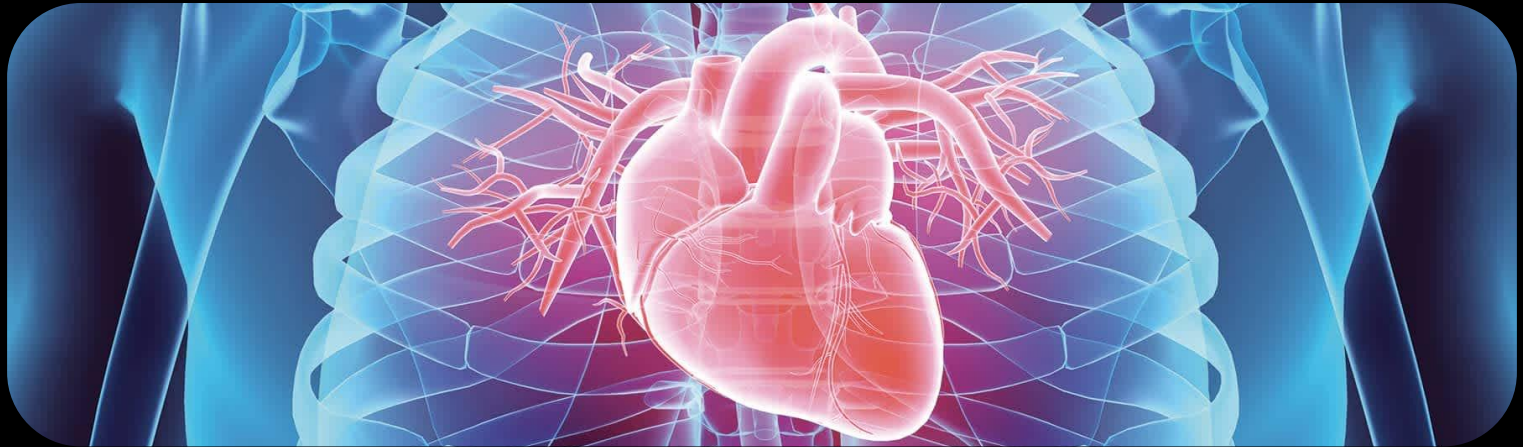


# STAT 479 Project

## Heart Diseases



**GOAL:**  
**Select meaningful features to predict if the patient has heart diseases by using their health data**

**1.**  
**Feature selection chooses useful features for predicting the target feature**

**2.**  
**Keeps useful features can make the model simpler and easier to understand**

**3.**  
**Improve model performance by reducing noise and redundancy**

**4.**  
**Help to decide which health factors are most related to heart disease.**

# About This File:

**Age:** Patients Age in years (Numeric)

**Sex:** Gender (Male : 1; Female : 0) (Nominal)

**cp:** Type of chest pain experienced by patient. This term categorized into 4 categories: 0 typical angina, 1 atypical angina, 2 non-anginal pain, 3 asymptomatic (Nominal)

**trestbps:** patient's level of blood pressure at resting mode in mm/HG (Numerical)

**chol:** Serum cholesterol in mg/dl (Numeric)

**fbs:** Blood sugar levels on fasting > 120 mg/dl represents as 1 in case of true and 0 as false (Nominal)

**restecg:** Result of electrocardiogram while at rest are represented in 3 distinct values: 0 : Normal 1: having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mV) 2: showing probable or definite left ventricular hypertrophy by Estes' criteria (Nominal)

**thalach:** Maximum heart rate achieved (Numeric)

**exang:** Angina induced by exercise 0 depicting NO 1 depicting Yes (Nominal)

**oldpeak:** Exercise induced ST-depression in relative with the state of rest (Numeric)

**slope:** ST segment measured in terms of slope during peak exercise: 0: up sloping; 1: flat; 2: down sloping (Nominal)

**ca:** The number of major vessels (0-3) (nominal)

**thal:** A blood disorder called thalassemia: 0: NULL 1: normal blood flow 2: fixed defect (no blood flow in some part of the heart) 3: reversible defect (a blood flow is observed but it is not normal) (nominal)

**target:** It is the target variable which we have to predict 1 means patient is suffering from heart disease and 0 means patient is normal.

# Preprocessing

## Target

Separate the target feature from other features:  
target is whether the person has heart disease.

All other columns are features that helps to predict the target

## Empty Data

Check for missing values before using any feature selection methods.

If there are missing values, the data might become biased, so remove them.

## Outliers

Check for outliers in variables because very large or unusual values might affect feature selection and lead to a biased result/interpretation.

## Standardization

Scale or standardize the numeric features to avoid variables with large numbers having a huge impact on the model.

# FILTER METHOD: CORRELATION SCREENING

**Using a significance level of 0.05,  
the selected features are:**

cp3, thal3, oldpeak, exang1, thalach,  
slope1, sex1, cp2, ca3, ca2, cp1,  
restecg2, trestbps, age, ca1

**Convert these features to its original  
name, the selected features are:**

cp, thal, oldpeak, exang, thalach,  
slope, sex, ca, restecg, trestbps, age

Test accuracy: 0.7963

Test AUC: 0.8310

```
> print(top_filter_results)
```

|    | feature  | correlation  | abs_correlation | p_value      |
|----|----------|--------------|-----------------|--------------|
| 5  | cp3      | 0.507695623  | 0.507695623     | 1.491354e-15 |
| 20 | thal3    | 0.494972383  | 0.494972383     | 9.465018e-15 |
| 13 | oldpeak  | 0.429695763  | 0.429695763     | 4.065089e-11 |
| 12 | exang1   | 0.420834111  | 0.420834111     | 1.109900e-10 |
| 11 | thalach  | -0.383170813 | 0.383170813     | 5.796946e-09 |
| 14 | slope1   | 0.346246973  | 0.346246973     | 1.773241e-07 |
| 2  | sex1     | 0.324764801  | 0.324764801     | 1.070416e-06 |
| 4  | cp2      | -0.311008172 | 0.311008172     | 3.155303e-06 |
| 18 | ca3      | 0.299761302  | 0.299761302     | 7.341336e-06 |
| 17 | ca2      | 0.252462460  | 0.252462460     | 1.769613e-04 |
| 3  | cp1      | -0.249325710 | 0.249325710     | 2.141667e-04 |
| 10 | restecg2 | 0.208169552  | 0.208169552     | 2.101743e-03 |
| 6  | trestbps | 0.191167968  | 0.191167968     | 4.812407e-03 |
| 1  | age      | 0.166262446  | 0.166262446     | 1.442820e-02 |
| 16 | ca1      | 0.161943419  | 0.161943419     | 1.721858e-02 |
| 7  | chol     | 0.126572574  | 0.126572574     | 6.332389e-02 |
| 15 | slope2   | 0.063150898  | 0.063150898     | 3.556625e-01 |
| 19 | thal2    | 0.022553892  | 0.022553892     | 7.417085e-01 |
| 9  | restecg1 | 0.008989884  | 0.008989884     | 8.954907e-01 |
| 8  | fbs1     | -0.001719124 | 0.001719124     | 9.799598e-01 |

# WHY?

1

These selected features showed a statistically significant relationship with the target feature.

In each selected features, at least one dummy variable had a p-value below 0.05. This means the association is unlikely to be due to random chance.

2

For example, cp and thal had very small p-values and relatively large correlations, so they look highly informative.

oldpeak, exang, thalach, and slope also stood out, which makes sense because they are closely related to exercise response and ECG-related heart stress patterns.

3

We don't want to select features like chol and fbs because their p-values were above 0.05, which means insignificance.

This means that these features does not have enough evidence to prove their significance to the target feature.

4

For example, oldpeak is important because it measures ST depression during exercise. This reflects how the heart responds under physical stress. A larger abnormal change can be a sign that the heart is not getting enough oxygen.

# EMBEDDED METHOD: LASSO

## R Output:

```
              s0
(Intercept) -0.20654780
age          .
sex1         0.52234810
cp1          .
cp2         -0.06964518
cp3          0.77161562
trestbps     0.27377586
chol         0.19955612
fbs1         .
restecg1     .
restecg2     0.14941515
thalach      -0.12557779
exang1       0.23301675
oldpeak      0.45006704
slope1       0.32200020
slope2       .
ca1          0.44190627
ca2          0.52984130
ca3          0.71638002
thal2        .
thal3        0.58485697
```

## Using Lasso, the selected features are:

```
sex1 = 0.5223
cp2 = -0.0696
cp3 = 0.7716
trestbps = 0.2738
chol = 0.1996
restecg2 = 0.1494
thalach = -0.1256
exang1 = 0.2330
oldpeak = 0.4501
slope1 = 0.3220
ca1 = 0.4419
ca2 = 0.5298
ca3 = 0.7164
thal3 = 0.5849
```

## Selected Features:

Sex, cp, trestbps, chol,  
restecg, thalach,  
exang, oldpeak, slope,  
ca, thal

Lasso test accuracy:  
0.7778

Lasso test AUC:  
0.8608



# WHY?

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## Why are these features selected?

The Lasso method with L1 penalty pushes less meaningful coefficients toward 0. After fitting the model, features with a coefficient that's not 0 are selected, and features with coefficient of 0 are not selected.

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## Interpret:

For example, cp is an important feature because chest pain type is closely related to heart problems. Different chest pain categories can reflect different levels of cardiac risk, so it makes sense that some cp dummy variables stayed in the model.

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## Why is Lasso useful?

Lasso is useful because it does not look at each variable one by one. Instead, it does feature selection and model fitting at the same time. It considers all variables together and keeps the ones that help the model most while shrinking weaker ones to zero. By using this method, we can reduce the chance of overfitting.

---

# Wrapper Method: LOCO

## Selected Features:

Using LOCO, the  
selected features are  
**thal, oldpeak, slope,  
restecg, and thalach**

LOCO test accuracy: 0.7222  
LOCO AUC: 0.8139

```
> print(loco_results_rounded)
  feature auc_without auc_drop acc_without
1    thal      0.7798   0.0341      0.7222
2 oldpeak      0.8068   0.0071      0.7407
3    slope      0.8082   0.0057      0.7407
4 restecg      0.8111   0.0028      0.7407
5  thalach      0.8125   0.0014      0.7222
6     fbs      0.8139   0.0000      0.7222
7 trestbps      0.8153  -0.0014      0.7778
8     sex      0.8153  -0.0014      0.7963
9     age      0.8153  -0.0014      0.7222
10    cp      0.8182  -0.0043      0.7778
11   chol      0.8196  -0.0057      0.7963
12  exang      0.8210  -0.0071      0.7407
13    ca      0.8693  -0.0554      0.7593

> selected_loco <- loco_results$feature[loco_results$auc_drop > 0]
> print(selected_loco)
[1] "thal"    "oldpeak" "slope"   "restecg" "thalach"
```

# WHY?

## Why LOCO?

In the LOCO method, the model is first trained using all original features.

Then, one feature is removed at a time, and the model is trained again to see how the test AUC changes.

If removing a feature causes the test AUC to decrease, that feature is considered important because it was helping the model make better predictions

## Interpretation:

Feature fbs were not selected by LOCO because its AUC drop is 0. This means that this feature is not having a significant impact to the model.

Features like trestbps, sex, age, cp, chol, exang, and ca have negative AUC drop, which means the model will perform better without them.

For example, when ca was removed, AUC increased to 0.8693.

## Interpretation:

Thal is important because it tells whether blood flow to the heart is normal or not. Abnormal blood flow might be due to possible heart defects, which is related to heart disease.

slope is important because it measures how the heart behaves under stress, so it can be useful for detecting abnormal heart function.

# Similarities/Differences:

All three methods are designed to reduce unnecessary variables and select the ones that carry important information. In all the methods, features connected to exercise response and heart condition are selected, such as oldpeak, thalach, thal, and slope.

The difference between these three methods is how they decide what features are important. The filter method, which in this case I used correlation screening method, looks at each feature one at a time, using coefficients and p-values to decide which feature is important. This means it checks the individual relationship between each feature and the response, but it does not consider how features work together.

Lasso adds an L1 penalty that shrinks weaker coefficients toward zero. Unlike correlation screening, Lasso considers all features together, so it captures more conditional relationships.

LOCO starts with the full model and removes one feature at a time to check how much the test AUC changes. Unlike the other two methods, LOCO measures the contribution of a feature to prediction.

# PROs and CONs

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## Correlation Screening:

Correlation screening is fast and easy to interpret, but it evaluates each feature separately. Because of this, it may select redundant features and ignore how features work together.

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## Lasso:

Lasso helps to reduce multicollinearity and reduce overfitting by capturing conditional relationships. However, it only captures linear relationships and might be unstable when features are highly correlated because it may keep one feature and remove another similar one.

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## LOCO:

LOCO removes one feature at a time and checks whether the AUC or accuracy gets worse, so it directly shows whether a feature adds useful information to the model. However, it is unstable when there are highly correlated features.

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# Summary:

Based on the results from the three feature selection methods, the most important features for predicting heart disease are thal, oldpeak, slope, and thalach, because these features were selected by correlation screening, Lasso, and LOCO method.

Other features such as sex, cp, trestbps, restecg, exang, and ca appeared in both correlation screening and Lasso, which suggests that these features may provide additional useful information even though they are not selected by LOCO.

This result is reasonable and connects to the goal because these are key features that can help predicting if the patient is likely to have heart disease or not.

I believe selecting these features will help create a model that is more accurate, easier to interpret, and better focused on the most meaningful signals related to heart disease.